

SESSION 3

Probability & Statistical Inference

Frequentist foundations and the Bayesian bridge

FROM SESSION 2

Tests, ANOVA and logistic regression

TODAY

Why inference works—and where it can mislead

TOWARD SESSION 6

Posterior computation and MCMC

Learning outcomes

By the end, you should be able to...

Explain probability, conditional probability, expectation and variance in statistical inference.

Distinguish a population parameter, estimator, estimate, sampling distribution and standard error.

Interpret confidence intervals and p-values correctly—and identify common misinterpretations.

Explain Type I/II errors, power, effect size and the role of sample size.

Describe key concerns in frequentist practice without dismissing the framework.

Explain prior, likelihood, posterior, credible intervals and posterior probabilities.

Compare frequentist and Bayesian answers to the same health question and explain why MCMC becomes necessary.

The inferential journey

A single map for the whole session

1 Probability

Model uncertainty

2 Sampling

Understand variability

3 Frequentist

Long-run calibration/
frequency

4 Bayesian

Update uncertainty

5 Computation

Approximate complex
posteriors

One health question throughout

Estimating hypertension prevalence

Observed: $x = 30$ hypertensive adults out of $n = 120$ sampled adults

FREQUENTIST QUESTION

How would an estimator/test behave over repeated samples if a specified parameter value were true?

BAYESIAN QUESTION

Given the observed data and a specified prior model, what uncertainty remains about prevalence p ?

Probability: the language of uncertainty

Core objects

Random variable

A numerical quantity whose value is generated by a stochastic process.

Example:

X = number hypertensive in n adults.

Probability distribution

Assigns probabilities to possible values or ranges. It is the mathematical model for uncertainty.

Parameter

A quantity indexing a distribution, e.g., prevalence p in a Binomial model.

Data model: $X \mid p \sim \text{Binomial}(n, p)$

Expectation and variance

Center and spread of a random variable

$$E(X) = \sum x \cdot P(X=x)$$

$$\text{Var}(X) = E[(X - E(X))^2]$$

EXPECTATION

The probability-weighted long-run average.

For $X \sim \text{Binomial}(n, p)$: $E(X) = np$.

VARIANCE

Quantifies dispersion around the expectation.

For Binomial: $\text{Var}(X) = np(1-p)$.

Conditional probability

Why “given” changes the question

$$P(A \mid B) = P(A \cap B) / P(B)$$

HEALTH EXAMPLE

$P(\text{Disease} \mid \text{Positive test})$: probability of disease among people who test positive.

NOT THE SAME

$P(\text{Positive test} \mid \text{Disease})$: sensitivity of the test.
Reversing the condition changes the probability.

Bayes' theorem begins with conditional probability

Evidence updates uncertainty

$$P(A \mid B) = P(B \mid A) P(A) / P(B)$$

$P(A)$: prior probability

$P(B \mid A)$: likelihood of the evidence under A

$P(A \mid B)$: updated/posterior probability after observing B

In diagnostic testing, prevalence can strongly affect $P(\text{Disease} \mid \text{Positive})$.

Population, sample, parameter, statistic

Do not blur these four ideas

Population

Target group we want to learn about.

Parameter

Fixed but unknown population quantity in frequentist inference; e.g., p .

Sample

Observed units drawn or selected from the population.

Statistic

Function of the observed sample; e.g., $\hat{p} = x/n$.

Estimator, estimate and bias

Theoretical distinction that matters

Estimator: $\hat{p} = X/n$

Estimate: observed $\hat{p} = 30/120 = 0.25$

UNBIASEDNESS

An estimator is unbiased when its expected value equals the target parameter: $E(\hat{p})=p$.

PRECISION

Even an unbiased estimator varies from sample to sample. Precision is tied to its sampling variance/SE.

Sampling distributions

The engine of frequentist inference

Imagine repeating the same sampling process many times.

Each sample produces a different statistic $\hat{p}$.

The distribution of those hypothetical $\hat{p}$ values is the sampling distribution.

Frequentist standard errors, confidence intervals and p-values are defined through this repeated-sampling logic.

$$SE(\hat{p}) \approx \sqrt{p(1-p) / n}$$

Law of Large Numbers vs Central Limit Theorem

Related, but not the same

LAW OF LARGE NUMBERS

As n grows, a sample average/proportion tends to get closer to its population target under appropriate conditions.

Think: consistency.

CENTRAL LIMIT THEOREM

After suitable centering/scaling, many estimators have an approximately Normal sampling distribution for sufficiently large n .

Think: shape for inference.

LLN → where the estimator goes CLT → how its sampling fluctuations behave

Frequentist inference: the central idea

Probability is attached to data/procedures under repeated sampling

The parameter is treated as fixed but unknown.

Before data are observed, the sample and statistics are random.

Procedures are evaluated by long-run operating characteristics: coverage, Type I error, power, bias, variance.

Inference asks what the observed data imply when viewed through those calibrated procedures.

Point estimation and standard error

Estimate + uncertainty

$$\hat{p} = x/n = 0.25$$

$$\text{Estimated SE}(\hat{p}) = \sqrt{[\hat{p}(1-\hat{p}) / n]}$$

POINT ESTIMATE

Best single-number summary from the sample.

STANDARD ERROR

Estimated sampling variability of the estimator—not the spread of individual observations.

Confidence intervals

A procedure with a long-run coverage guarantee

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

A 95% confidence procedure is constructed to cover the true parameter about 95% of the time under repeated sampling, when assumptions hold.

Once the interval is computed, the frequentist parameter is not random.

The interval's width reflects sample size, variability and confidence level.

A confidence interval is NOT a credible interval

Same-looking numbers; different probability statements

95% CONFIDENCE INTERVAL

Correct idea: the method has 95% long-run coverage under the assumed data-generating process.

Avoid: “There is a 95% probability p is inside this realized interval.”

95% BAYESIAN CREDIBLE INTERVAL

Conditional on the model and prior, 95% of the posterior probability for p lies in the interval.

This directly assigns probability to parameter uncertainty.

Hypothesis testing

A model-checking calculation under H_0

$$H_0: p = p_0 \quad H_1: p \neq p_0$$

Choose a test statistic whose distribution under H_0 is known or approximated.

Calculate the observed statistic.

The p-value measures how incompatible the observed statistic—or more extreme values—are with H_0 under the test model.

Decision rules such as $\alpha = 0.05$ control long-run Type I error under stated assumptions.

The p-value: precise definition

Conditional on H_0 and the statistical model

p-value = $P(\text{test statistic as or more extreme than observed} \mid H_0)$

IT DOES TELL US

How surprising the observed test statistic is under H_0 , according to the chosen test and assumptions.

IT DOES NOT TELL US

$P(H_0 \text{ is true} \mid \text{data})$, the probability results are “due to chance,” or the magnitude/importance of an effect.

Type I error, Type II error and power

Decision errors are defined across repeated studies

TYPE I ERROR (α)

Reject H_0 when H_0 is true. α is the long-run false-positive rate of the decision rule.

TYPE II ERROR (β)

Fail to reject H_0 when a specified alternative is true. This is a false negative.

POWER ($1-\beta$)

Probability of rejecting H_0 for a specified true effect. Depends on effect size, n , variability and α .

Power is not a property of the p-value; it is a property of a study design under a specified alternative.

Effect size, precision and statistical significance

Three different ideas

EFFECT SIZE

How large is the difference or association?
Clinical relevance belongs here.

PRECISION

How uncertain is the estimate?
Often summarized with SE or interval width.

SIGNIFICANCE

How incompatible are data with H_0 under the test model? Often summarized by p-value.

Large n can make tiny effects statistically significant; small n can leave important effects uncertain.

Multiple testing and selective inference

Why repeated testing changes false-positive behavior

If many null hypotheses are tested, the chance of at least one small p-value increases.

Post-hoc outcome switching, subgroup searching and repeated model selection can invalidate naive p-value interpretations.

Pre-specification, multiplicity adjustment, transparent reporting and replication are important safeguards.

For m independent tests at level α : $P(\text{at least one false positive}) = 1 - (1-\alpha)^m$

Concerns with frequentist practice

Important critiques—without caricature

Dichotomizing evidence into “significant / not significant” at $p < 0.05$ discards information.

p-values and CIs are frequently misinterpreted.

Results can be strongly sample-size dependent.

Optional analyses and multiple comparisons require careful error control.

Standard frequentist output does not directly provide $P(\text{parameter} > \text{threshold} \mid \text{data})$.

But frequentist methods offer powerful design-based and long-run guarantees when used appropriately.

Transition: change the inferential target

From repeated-sampling behavior to uncertainty about parameters

FREQUENTIST LENS

“If we repeatedly generated data under this model, how would this procedure behave?”

Calibration: coverage, α , power, bias, variance.

BAYESIAN LENS

“Given the observed data and prior model, what values of the parameter remain plausible?”

Output: posterior distribution and posterior probabilities.

Bayesian inference: the mathematical core

Famous Bayesian Inference Quote

A Bayesian is one who, vaguely expecting to see a horse and catching a glimpse of a donkey, strongly concludes he has seen a mule. (Senn, 1997)



Bayesian inference: the mathematical core

Probability is used to quantify uncertainty about unknown parameters

Posterior $\propto$ Likelihood $\times$ Prior

$$p(\theta \mid y) = p(y \mid \theta) p(\theta) / p(y)$$

θ : unknown parameter(s)

y : observed data

$p(\theta)$: prior distribution

$p(y \mid \theta)$: likelihood

$p(\theta \mid y)$: posterior distribution

Prior distributions

Make assumptions explicit

WEAKLY INFORMATIVE

Constrains implausible extremes while allowing the data substantial influence.

INFORMATIVE

Represents substantive prior evidence or expert knowledge. Requires careful justification.

SENSITIVITY ANALYSIS

Repeat inference under reasonable alternative priors to assess how conclusions depend on prior assumptions.

A prior is part of the model—not an optional decoration.

Likelihood

How the observed data inform parameters

$$L(p; x, n) \propto p^x (1-p)^{n-x}$$

For Binomial data, the likelihood is a function of p with the observed x and n held fixed.

Values of p that make the observed data more probable receive greater likelihood support.

The likelihood is not a probability distribution for p by itself; Bayesian inference combines it with a prior.

Beta–Binomial updating

A closed-form Bayesian health example

Prior: $p \sim \text{Beta}(a, b)$ Data: $X | p \sim \text{Binomial}(n, p)$

Posterior: $p | x \sim \text{Beta}(a+x, b+n-x)$

INTUITION

Observed successes add to a ; observed failures add to b .

WHY IT MATTERS

We can see Bayesian updating analytically before introducing MCMC.

Prior effective sample size

A useful Beta-prior interpretation

For Beta(a,b): prior mean = $a/(a+b)$ prior concentration $\approx a+b$

Larger $a+b$ generally produces a more concentrated prior around its mean.

The “effective sample size” interpretation is heuristic and parameterization-dependent, but useful pedagogically in the Beta-Binomial case.

As n grows, a fixed prior usually contributes less relative influence to the posterior.

Posterior summaries

The posterior is the primary object

LOCATION

Posterior mean, median or mode summarize central tendency.

UNCERTAINTY

Credible intervals summarize posterior probability over parameter ranges.

DECISION-RELEVANT PROBABILITY

Compute quantities such as $P(p > 0.20 \mid \text{data})$
or
 $P(\text{effect} > \text{clinically meaningful threshold} \mid \text{data})$.

Frequentist CI vs Bayesian credible interval

A direct comparison

Feature	Frequentist 95% CI	Bayesian 95% credible interval
Parameter	Fixed, unknown	Uncertainty represented by posterior
Probability statement	About procedure coverage	About parameter under model + prior
Depends on	Sampling model/procedure	Likelihood + prior
Direct $P(\theta > c \mid \text{data})$?	Not from CI alone	Yes, from posterior

Same hypertension data, two analyses

$x = 30$ of $n = 120$

FREQUENTIST

Estimate $\hat{p} = 0.25$

Construct a 95% confidence interval

Optionally test $H_0: p = 0.20$

Report estimate, interval and p-value

Interpret through repeated-sampling calibration.

BAYESIAN

Specify a prior, e.g. $p \sim \text{Beta}(2,8)$

Combine with Binomial likelihood

Obtain posterior $\text{Beta}(32,98)$

Report posterior mean/credible interval

Compute $P(p > 0.20 \mid \text{data, model, prior})$.

Bayesian inference also has concerns

Different framework, different responsibilities

Prior choice can matter, especially with limited or weakly informative data.

Posterior conclusions are conditional on the likelihood, prior and model structure.

Prior-data conflict should be investigated, not hidden.

Sensitivity analysis and posterior predictive checking are important.

Complex models require computation and convergence diagnostics.

Bayesian methods do not repair confounding, selection bias, measurement error or poor design.

Why computation becomes necessary

Closed-form posteriors are the exception in realistic models

Beta-Binomial updating is analytically convenient because the prior and likelihood are conjugate.

Bayesian logistic regression, multilevel models and many modern health models produce posteriors that cannot be normalized or summarized by simple algebra.

We therefore approximate posterior expectations, intervals and probabilities numerically.

Complex posterior → numerical approximation → MCMC / related algorithms

Bridge to Session 6: MCMC

From posterior theory to posterior computation

1 TARGET

Define the posterior distribution implied by likelihood + priors.

2 SAMPLE

Construct a Markov chain whose long-run distribution is the target posterior.

3 DIAGNOSE & SUMMARIZE

Assess convergence/mixing, then estimate posterior means, intervals and probabilities.

Session 3: What is a posterior? **Session 6: How do we compute one?**

Method-selection discussion

Ask the scientific question before choosing the inferential framework

Do I need long-run error control or repeated-sampling calibration?

Do I need direct probability statements about parameters or clinically meaningful thresholds?

Is credible prior information available—and can I justify it?

How sensitive are conclusions to assumptions, model choices and missing data?

What will the audience, discipline, regulator or decision-maker need to interpret?

Key takeaways

Five ideas to carry forward

Probability models connect uncertain data to scientific parameters.

Frequentist inference is built around repeated-sampling properties of procedures.

p-values, confidence intervals, effect sizes and clinical importance answer different questions.

Bayesian inference combines prior information and likelihood to produce a posterior distribution.

MCMC becomes necessary when the posterior is too complex to calculate directly.

Exit ticket

Explain each in one sentence

What does a p-value condition on?

Why is a 95% confidence interval not a 95% credible interval?

What are prior, likelihood and posterior?

Give one concern about frequentist practice and one concern about Bayesian practice.

Why will we need MCMC in Session 6?